



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ENGINEERING GRAPHICS AND DESIGN P2

MAY/JUNE 2025

MARKING GUIDELINES

MARKS: 100

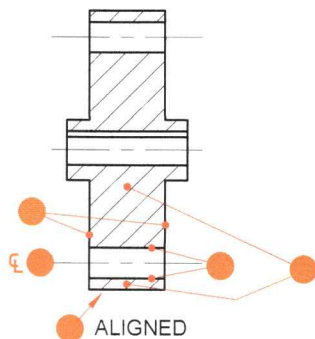
These marking guidelines consist of 7 pages.

ANSWERS

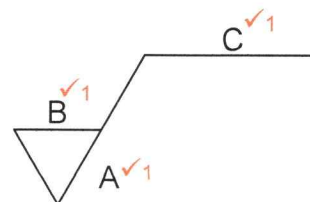
1	SAMANTHA	1
2	CLAMPIT TOOLS CC	1
3	AutoCAD 2024	1
4	THIRD-ANGLE ORTHOGRAPHIC / TAOP	1
5	MILD STEEL	1
6	2	1
7	860°	1
8	TOP VIEW	1
9	DETAIL T	1
10	A: R2 B: 9 C: R7 D: Ø122 E:70	5
11	ROUNDING	1
12	BOLT	1
13	R63	1
14	REVOLVED SECTION	1
15	PART SECTION	1
16	SITE WELD / FIELD WELD	1
17	25 = LENGTH 75 = PITCH	2
18	MAXIMUM: 40,23 MINIMUM: 39,9	2
19	See below	3
20		3
TOTAL		30

ANSWER 19

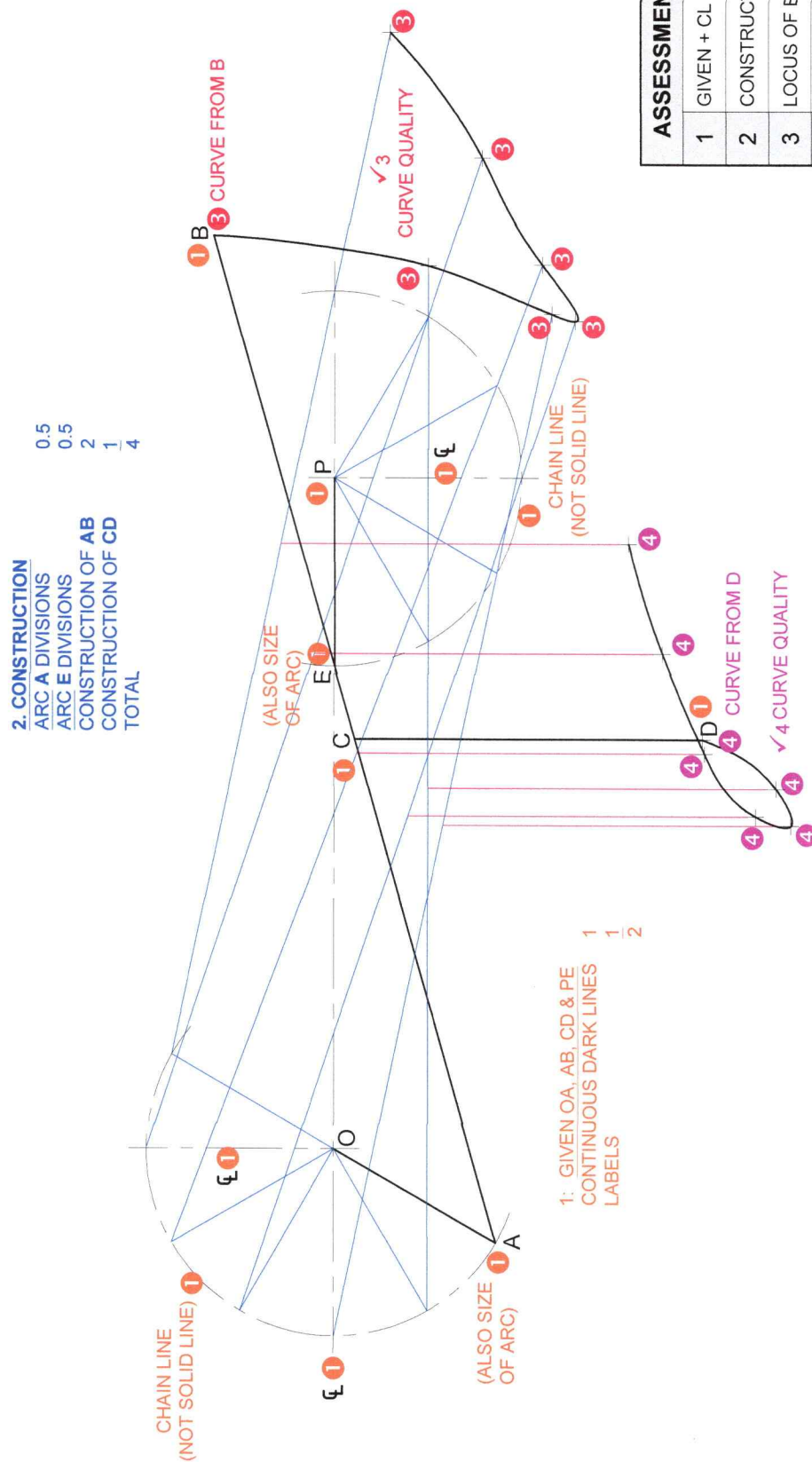
 FREEHAND



ANSWER 20



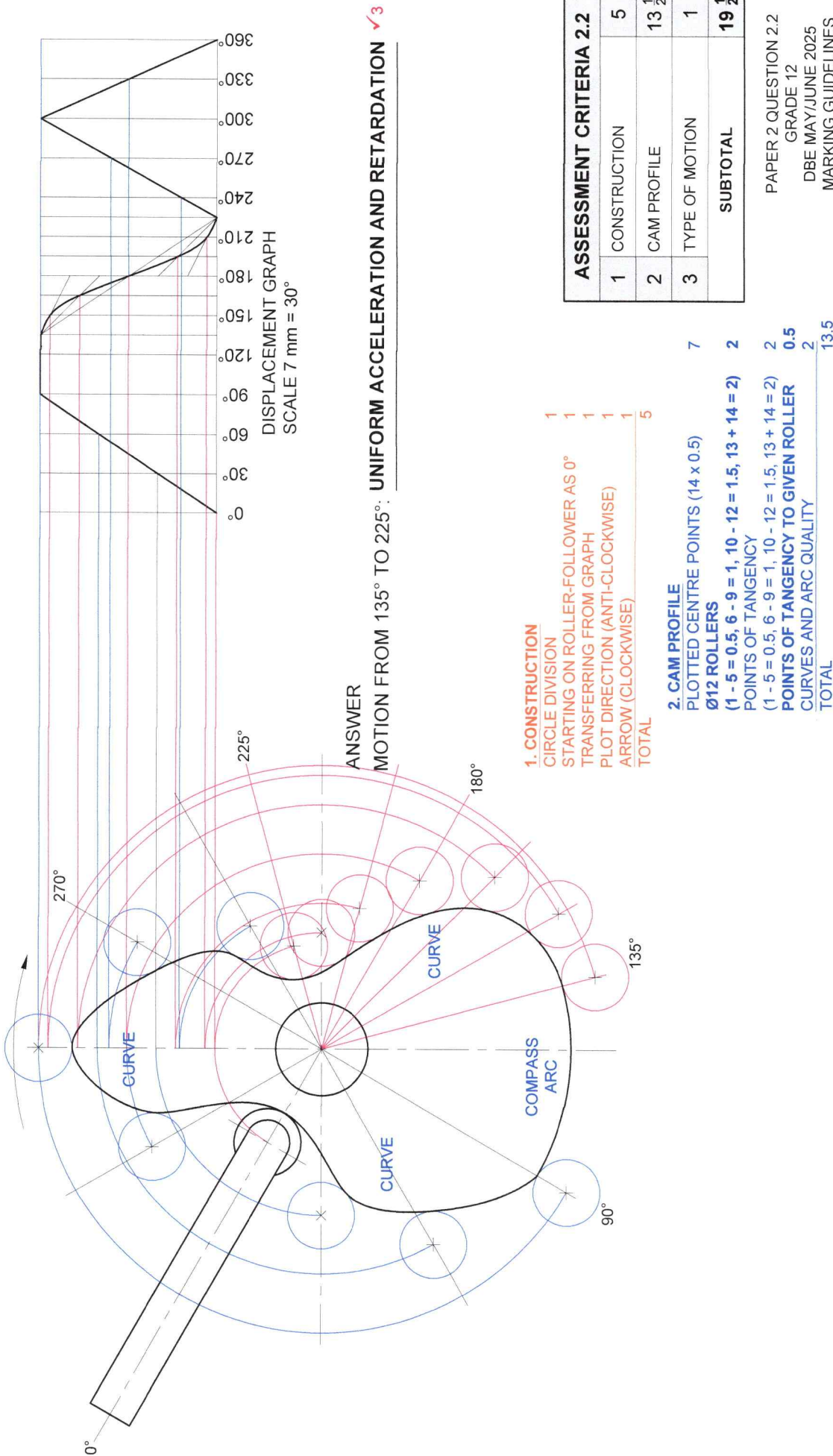
PAPER 2 QUESTION 1
GRADE 12
DBE MAY/JUNE 2025
MARKING GUIDELINES



ASSESSMENT CRITERIA 2.1		
1	GIVEN + CL	$7\frac{1}{2}$
2	CONSTRUCTION	4
3	LOCUS OF B	$4\frac{1}{2}$
4	LOCUS OF D	$4\frac{1}{2}$
SUBTOTAL		$20\frac{1}{2}$

PAPER 2 QUESTION 2.1
GRADE 12

DBE MAY/JUNE 2025
MARKING GUIDELINES



ASSESSMENT CRITERIA 2.2			
1	CONSTRUCTION	5	
2	CAM PROFILE	13.5	
3	TYPE OF MOTION	1	
SUBTOTAL		19.1	

PAPER 2 QUESTION 2.2
GRADE 12
DBE MAY/JUNE 2025
MARKING GUIDELINES

The diagram shows an isometric view of a mechanical component. It features a base with a hexagonal cutout, a central rectangular block, and a top section with a curved profile. Marking points are indicated by colored circles: red (3), orange (6), pink (4), green (5), and blue (2). Construction lines include a green circle with center 'CL' and radius '5', and a pink line labeled 'CIRCLE CONSTRUCTION'. Sectioned surfaces are indicated by 'H' and hatching. Two auxiliary views are shown at the bottom left: a front view (labeled 1) and a side view (labeled 1).

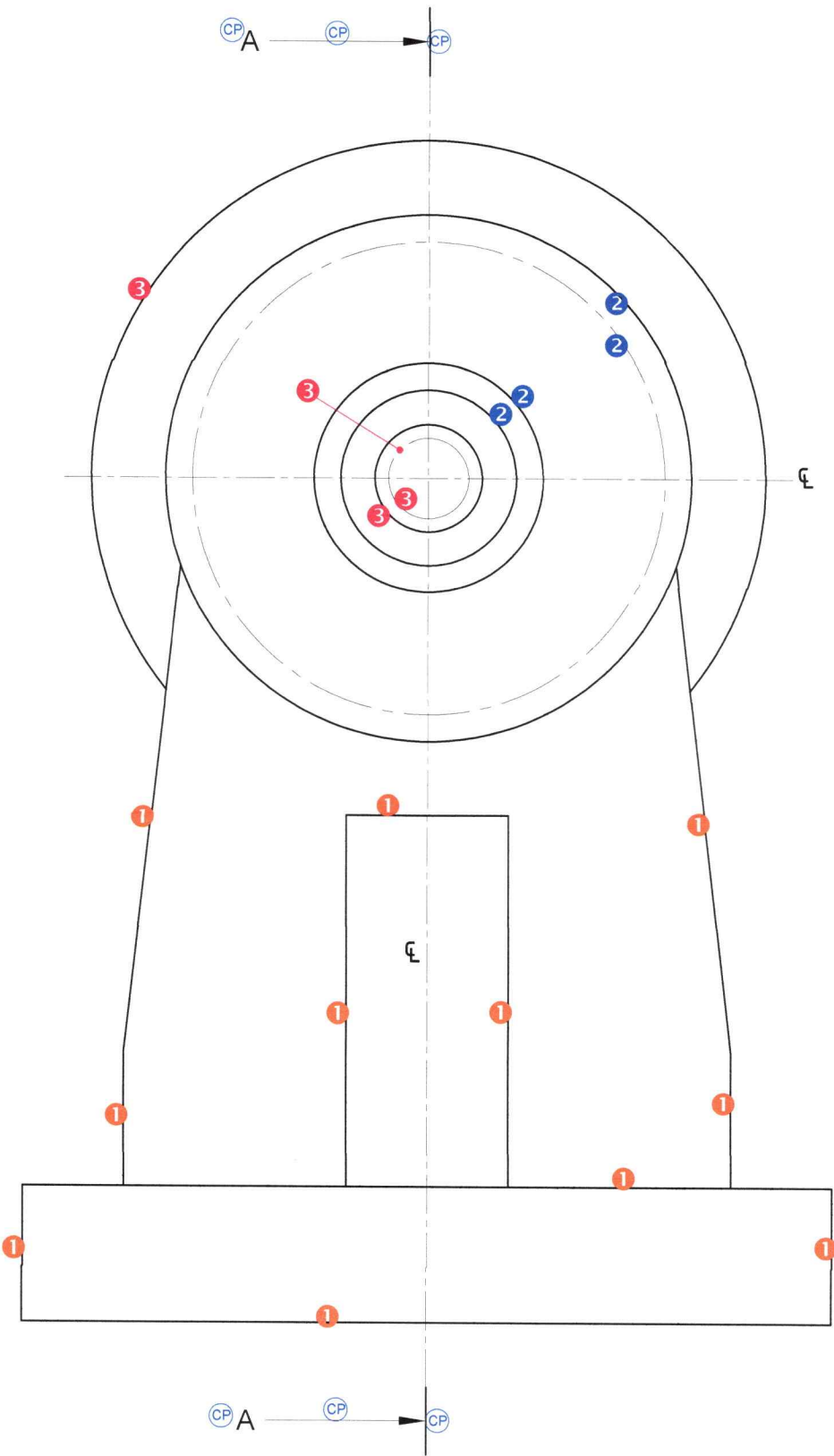
ASSESSMENT CRITERIA		
1	PLACING/ORIENTATION + AUX. VIEWS	2
2	FRONT PORTION	16 $\frac{1}{2}$
3	MIDDLE + REAR PORTION	7
4	HEXAGON	4 $\frac{1}{2}$
5	CIRCLE + CL	5
6	SECTIONED SURFACE + HATCHING	5
TOTAL		40

PAPER 2 QUESTION 3

GRADE 12

DBE MAY/JUNE 2025

MARKING GUIDELINE

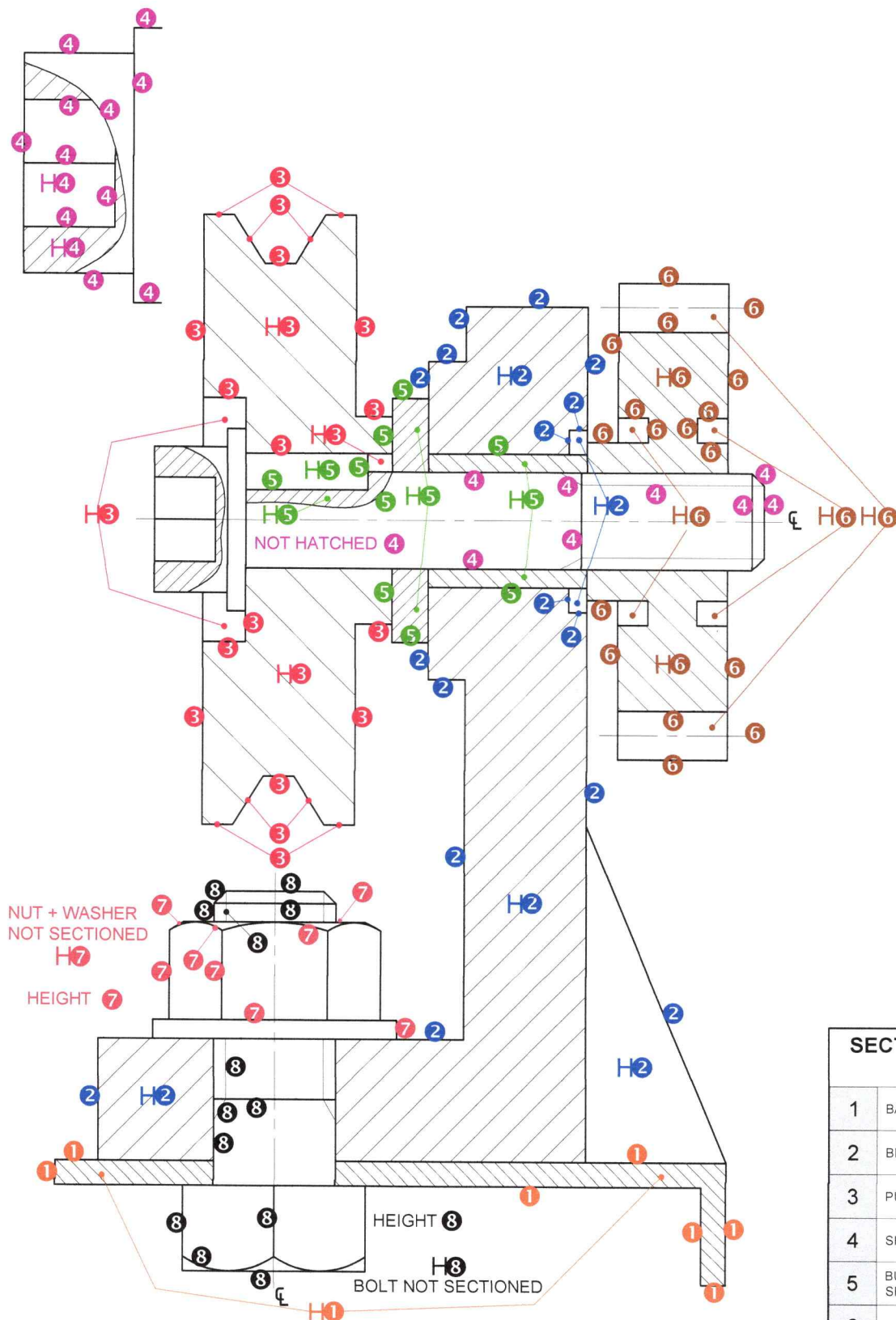


1. CENTRE LINES (CL)
SANS COMPLIANT
CENTRE LINES (1 - 2 = 0.5; 3 - 4 = 1)

1
1
2

RIGHT VIEW		
1	BRACKET + BASE PLATE	5 ¹ / ₂
2	GEAR + CL	2
3	PULLEY + SHAFT	2
SUBTOTAL		9 ¹ / ₂

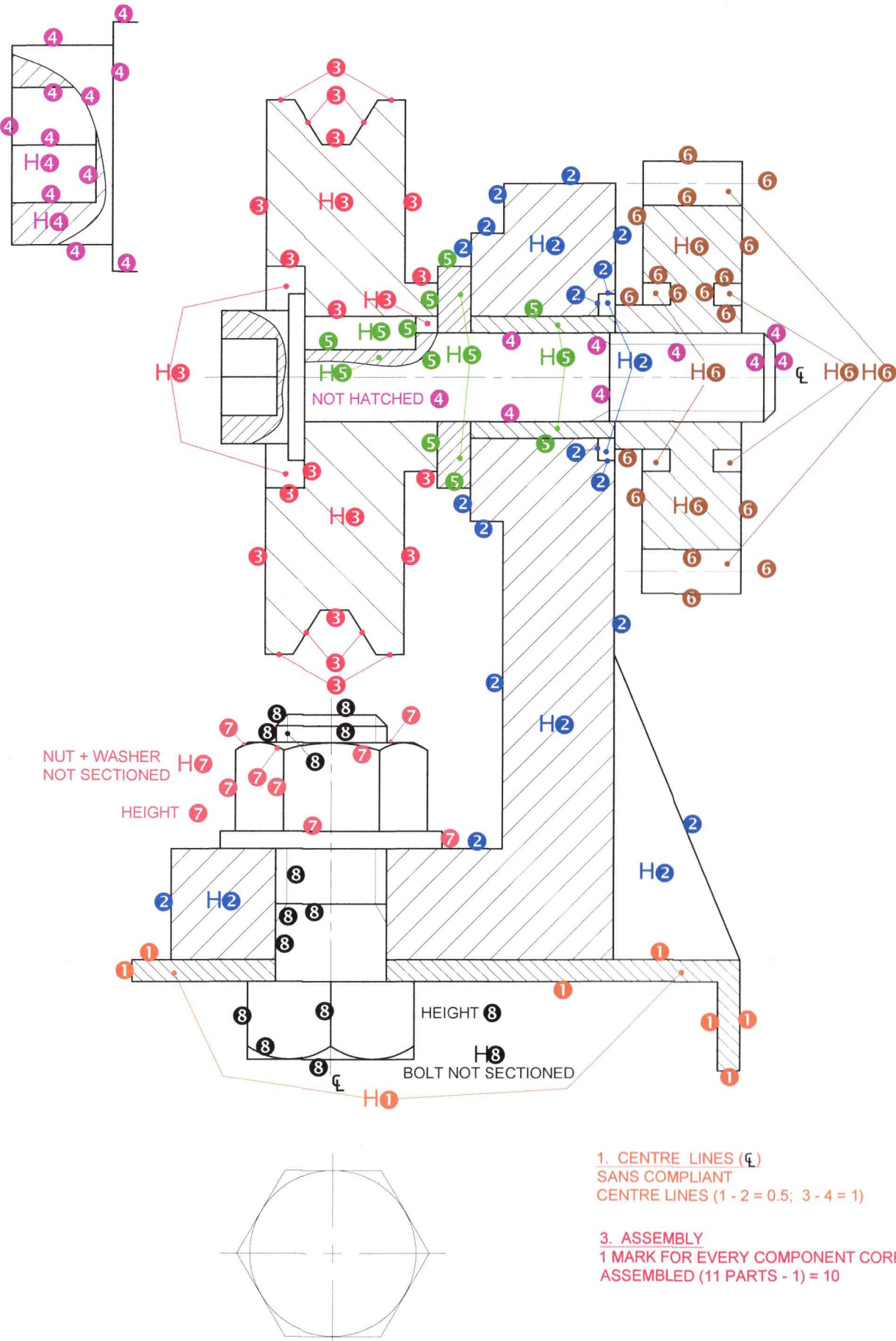
PAPER 2 QUESTION 4
GRADE 12
DBE MAY/JUNE 2025
MARKING GUIDELINE



SECTIONAL FRONT VIEW

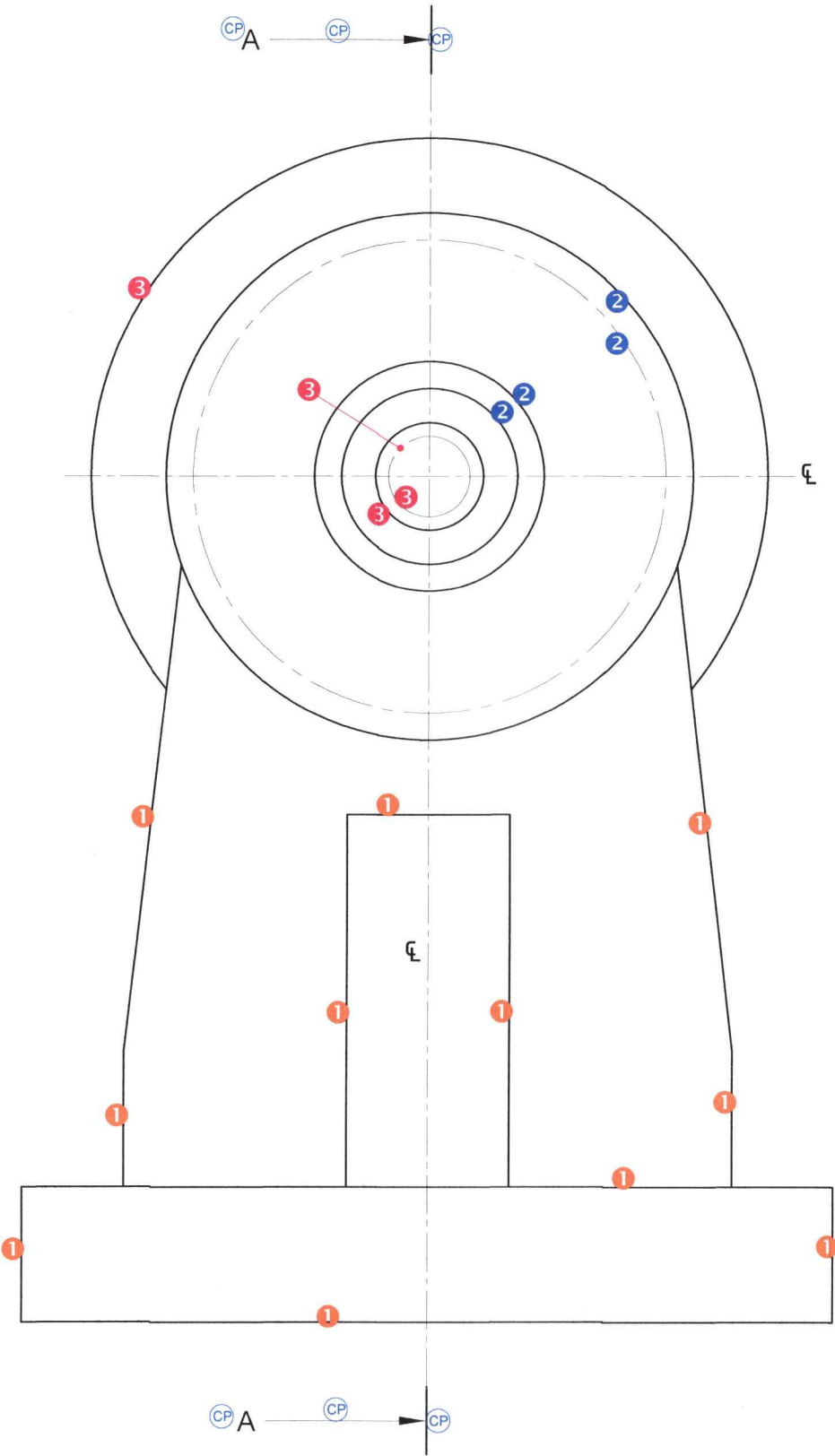
1	BASE	4
2	BRACKET + RIB	10 $\frac{1}{2}$
3	PULLEY	10
4	SHAFT	11
5	BUSH + KEY + SPACER	6 $\frac{1}{2}$
6	SPUR GEAR	11
7	M20 NUT + WASHER	5
8	M20 BOLT	7 $\frac{1}{2}$
SUBTOTAL		65 $\frac{1}{2}$

PAPER 2 QUESTION 4
GRADE 12
DBE MAY/JUNE 2025
MARKING GUIDELINE



1. CENTRE LINES (CL)
SANS COMPLIANT
CENTRE LINES (1 - 2 = 0.5; 3 - 4 = 1) 1 1/2

3. ASSEMBLY
1 MARK FOR EVERY COMPONENT CORRECTLY
ASSEMBLED (11 PARTS - 1) = 10



FOR OFFICIAL USE ONLY	
INCORRECT ORTHOGRAPHIC PROJECTION	
INCORRECT OVERALL SCALE	
INCORRECT HATCHING	
PARTS NOT ASSEMBLED	
TOTAL PENALTIES (-)	

ASSESSMENT CRITERIA					
RIGHT VIEW					
		POSSIBLE	OBTAINED	SIGN	MODERATED
1	BRACKET + BASE PLATE	5 1/2			
2	GEAR + CL	2			
3	PULLEY + SHAFT	2			
SUBTOTAL		9 1/2			
SECTIONAL FRONT VIEW					
1	BASE	4			
2	BRACKET + RIB	10 1/2			
3	PULLEY	10			
4	SHAFT	11			
5	BUSH + KEY + SPACER	6 1/2			
6	SPUR GEAR	11			
7	M20 NUT + WASHER	5			
8	M20 BOLT	7 1/2			
SUBTOTAL		65 1/2			
GENERAL					
1	CENTRE LINES	2			
2	CUTTING PLANE	3			
3	ASSEMBLY	10			
SUBTOTAL		15			
TOTAL		90			
PENALTIES (-)					
GRAND TOTAL					